INTEGER WORD PROBLEMS

Name: ..... **Date:** .....

Write an integer expression for each problem (use negative numbers for below, down, spent and lost), work it out, and answer in a sentence with a unit.

1. The temperature at 6 a.m. was −4 degrees. By noon it had risen 11 degrees. What was the temperature at noon?

2. A diver is at −12 meters (12 m below the surface). She goes down another 8 meters. What is her position now?

3. Leo has $15 in his account. He spends $20. What is his balance?

4. A football team gained 7 yards, then lost 12 yards, then gained 4 yards. What was the total change?

5. A town was 3 degrees at midnight and the temperature dropped 5 degrees each hour for 3 hours. What was it at 3 a.m.?

6. An elevator starts on floor 6, goes down 9 floors to the parking level, then up 4. Which floor is it on? (Floors below ground are negative.)

Worked example: "It was −6 degrees and warmed up by 10 degrees" is −6 + 10 = 4 degrees. "A submarine at −30 m rises 12 m" is −30 + 12 = −18 m: still below the surface.

INTEGER WORD PROBLEMS ANSWER KEY

Name: ..... **Date:** .....

Answer key: one integer expression and its value for each problem.

1. The temperature at 6 a.m. was −4 degrees. By noon it had risen 11 degrees. What was the temperature at noon?

−4 + 11 = 7 degrees

2. A diver is at −12 meters (12 m below the surface). She goes down another 8 meters. What is her position now?

−12 − 8 = −20 meters

3. Leo has $15 in his account. He spends $20. What is his balance?

15 − 20 = −5 dollars (he owes $5)

4. A football team gained 7 yards, then lost 12 yards, then gained 4 yards. What was the total change?

7 + (−12) + 4 = −1 yard

5. A town was 3 degrees at midnight and the temperature dropped 5 degrees each hour for 3 hours. What was it at 3 a.m.?

3 + 3 × (−5) = −12 degrees

6. An elevator starts on floor 6, goes down 9 floors to the parking level, then up 4. Which floor is it on? (Floors below ground are negative.)

6 − 9 + 4 = 1, floor 1

Worked example: "It was −6 degrees and warmed up by 10 degrees" is −6 + 10 = 4 degrees. "A submarine at −30 m rises 12 m" is −30 + 12 = −18 m: still below the surface.

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